

TRIPS AND PHARMACEUTICAL PATENTS IN DJIBOUTI: AN ANT ANALYSIS OF SOCIO-LEGAL OBJECTS

EMILIE CLOATRE
University of Nottingham, UK

ABSTRACT

This article discusses the complexity inherent in the relationship between written law and social action. It uses actor-network theory ('ANT') to amplify this complexity and considers its value both in understanding socio-legal objects and, more broadly, to socio-legal studies itself. The article uses a case study of the role of the Trade Related Intellectual Property agreement ('TRIPS') regarding pharmaceutical patents in a 'least developed country', Djibouti. The study uses this pharmaceutical example to argue the insights offered by ANT, conceptualizing socio-legal objects, beyond comparable approaches such as implementation studies. It also offers a different, more compelling set of understandings than that which appears in the more standard texts on TRIPS.

KEY WORDS

actor-network theory; development; methodology; pharmaceutical patents; TRIPS

INTRODUCTION

THIS ARTICLE PRESENTS an empirical case study of the impact of the Trade Related Intellectual Property agreement (TRIPS) and pharmaceutical patents on health. This study, carried out in the Republic of Djibouti during 2004, also illustrates the potential relevance and limits of actor-network theory (ANT) as a method to study the action of socio-legal objects – understood as objects with a legal origin/dimension studied in their social action, through networks and connections, as exemplified throughout the article.¹

This inquiry is situated within an academic debate that has remained essentially doctrinal (Nogues, 1993; Rozek and Berkowitz, 1998; Correa, 2000; Otten, 2000; Velasquez, 2000). This example focuses on a type of country seldom studied and illustrates the complex linkages between the written creation of legal objects and their action in various networks. Although ANT remains rarely used in socio-legal research, it appeared relevant to this particular case study for several reasons. First, the research undertaken was in itself quite closely related to some aspects of science and technology studies (STS), from which ANT originated. Disease, drugs and pharmaceutical research are all areas discussed in STS (including early ANT writings; (Latour, 1979, 1987, 1993) and in the area of pharmaceutical patents. This explains how, in terms of research process and rationale, ANT appeared as beneficial to understanding some of the dynamics occurring in this field. At the same time, and although ANT is gaining prominence in socio-legal studies (Tully, 2006), its potential relevance to this field still needs to be tested, proven, and potentially improved. This case study appeared a feasible and appropriate opportunity to do so. As well as its thematic links with prior ANT writings, the case study features a relatively small network, compatible with the ethnographic approach that ANT requires.

The article begins by explaining key concepts and methods used in this research; the case study is then located within existing debates on TRIPS and pharmaceuticals; two key aspects of findings are introduced and discussed: the contrast between expectation and action of TRIPS and the action of pharmaceutical patents in Djibouti despite their lack of official existence. Finally, the article discusses the benefits and limits of ANT when applied to socio-legal analysis.

SOME KEY CONCEPTS OF ACTOR-NETWORK THEORY

ANT was developed in STS, to analyse how technologies work in society (Latour, 1993). It stresses the need to understand associations, connections and networks in order to seize the modes of action of relevant objects. It also shifts the focus of social analysis from humans to a broader range of actors and allows various types of non-human objects to be brought to the centre of social analysis.

ANT challenges other sociological approaches by substituting the stable idea of 'society' with the existence of a multiplicity of social orderings generated by sets of mobile associations (Latour, 2005). It stresses the need to consider how sets of fluid relationships create order and power by their existence and interdependence, while constantly shaping any object of enquiry (Law, 1992). The very notion of 'actor-network', understood as 'the summing up of interactions through various kinds of devices, inscriptions, forms etc, into a very local, very practical, very tiny locus' (Latour, 1999: 16), reflects the idea that any entity needs to be understood as the complementarity of sets of connections, as multi-dimensional and likely to be modified if any connection is challenged.

The emphasis in ANT on co-dependency challenges the ruptures and dichotomies accepted in many sociological analyses, preferring instead a symmetrical analysis of objects and networks (Latour, 1987, 1993; Law, 1999b). This idea originates in criticisms of the divide between nature and society, perceived in ANT as being continuous with each other, rather than as two different spheres (Latour, 1993), which is maybe more central for science studies than socio-legal research. More relevant to this study, ANT questions the divide between humans and non-humans (Law, 1992). The claim made is that, *although humans are morally and ethically different from non-humans, they are not the only determinants of social phenomena* (Latour, 1993). This idea, translated as ‘symmetry’ and ‘heterogeneity’, means that any entity is understood, regardless of its nature, as a potential actor within complex networks (Law, 1992). Humans no longer can be seen as the ‘centre’ or the ‘origin’ of action, and any social action is the result and expression of multiple interactions between various entities. We will see later how this can allow us to *discuss the action of ‘patents’ rather than, for example, the action of ‘patent holders’*. We will discuss the *advantages and limitations* of this approach in conclusion.

A significant part of ANT literature has analysed *how technologies express their multiple dimensions through various networks, potentially in ways contrasting with their initial purpose*. ANT has therefore developed significant expertise in demonstrating the life that technologies can take different from what their creators had designed them to be.

It is useful to refer to a classic example of ANT studies in order to seize the overlap between the aims of this research and those of researchers in ANT. A useful example is that of the *waterpump studied by de Laet and Mol (2000)*. In their research, they analyse the set of transformations this object undertook when transferred from its place of creation to Zimbabwean villages. They demonstrate the set of connections that makes it transform, and the networks that it will, in turn, modify. *Their focus is on that particular object and its social connections, and in understanding how it takes a life of its own away from its creator’s intentions*. The similarities between their aims in this research, and the purpose of the case study analysed in this article – and possibly of other socio-legal analyses – are clear: *in this case study, the focus was on the modes of action of TRIPS away from the WTO and into Djibouti, and the life that it – and pharmaceutical patents – could take through a particular set of social conditions and connections*.

Empirically, the role of researchers in ANT becomes to highlight the connections that make up their object of study, and to illustrate through these its modes of social action. ANT’s emphasis on the need to remain ‘modest’ (Law, 1994), and to consider actors on the ground as being most able to describe the world as it really is for them, means that ANT has been associated with ethnographic methods *valued for their flexibility and reflexivity* (Latour, 2005). In relation to socio-legal research, ANT offers a strategy to follow socio-legal objects in their various expressions, through the networks they ‘should’ influence, as well as through their unexpected and more silent expression.

In this research, data collection was based on in-depth interviews with 25 informants from the public health and trade and industry areas in Djibouti. These interviews were complemented by document analysis and observation. The research process was reflexive, and themes for investigation emerged in an inductive way, following actors and emerging connections throughout the course of this fieldwork. In doing so, the empirical guidance provided by ANT was essential. It offered a constant reminder of the need to avoid pre-established assumptions about what should or should not matter in this case study. It was important in this example when working in a country on which little had been written, and useful in a field where many statements have gone increasingly unquestioned. Clearly, this is something that cannot be attributed only to ANT, but should follow rigorous empirical research. Nevertheless, the insistence of ANT on 'following the actors', 'looking down' into networks and avoiding the temptation to draw frameworks or patterns that are not readily apparent provided additional incentives to remain empirically grounded.

In terms of theory and method, however, this research also opened the door to testing the relevance of ANT to socio-legal studies. In many respects, the potential overlapping of ANT and some socio-legal studies was seen as part of the attractiveness of this approach for the field, and in this case study, once again, this overlap was particularly clear in terms of aims and of themes. Concerns with comparing expectations and actions, or with understanding the reshaping of various objects, are common to ANT and socio-legal research, and this research aimed to understand, in part, how (and if) the methods used in ANT when analysing these questions could provide new or additional tools for socio-legal researchers.

TRIPS, PHARMACEUTICAL PATENTS AND DEVELOPING COUNTRIES: EXISTING DEBATES

The potential impact of TRIPS² and pharmaceutical patents on health in developing countries has been highly controversial (Nogues, 1993; Velasquez, 2000; Abbott, 2005). The profusion of academic texts on the issue makes it unnecessary to recap in detail the substance of the debate in this article. Only some specific points need to be made.

Until 1995, various countries did not protect pharmaceutical patents. This choice was often based on development policies, and frequently understood as triggering a local market in generic medicines (e.g. Scherer and Watal, 2002).³ Since the signature of TRIPS in 1995, however, every member state of the WTO is required to offer patent protection for pharmaceutical products (Art.27(1)). The deadline for this obligation varies according to the development status of the country (Art. 65 and 66). This is the result of intense debate between countries that host a strong research-based pharmaceutical industry, and other states – often framed as a 'North/South' debate (Drahos, 2002). TRIPS has been criticized, since there have been fears that the increased

effectiveness of pharmaceutical patents worldwide might result in generic medicines becoming less available, thereby increasing the price of medication (Muennich, 2000; Velasquez, 2000; Abbott, 2005). It is important to note that the effects of TRIPS have been fundamentally respecified by various TRIPS council decisions and WTO dispute resolutions since its adoption, including the Doha Declaration on Public Health and the TRIPS Council declaration of August 2003, thus reclarifying some aspects of it (Abbott, 2002; Matthews, 2004b). However, for the purpose of this article, it is unnecessary to discuss these. What will be relevant here are the basic first steps taken by Djibouti in trying to implement a nascent pharmaceutical patents system, the detail of which was still unspecified at the time of fieldwork. This will be illustrated later.

In spite of being extensive, literature on TRIPS presents limits that this research aims to address. The debate on TRIPS and 'developing states' essentially centres on specific countries – states that host a strong generic pharmaceutical industry, and that had made a rational development policy choice of excluding pharmaceuticals from their field of patentability (Henderson, 1997; Braune and Menezes, 1998; Lanjouw, 1998; Mayer, 1998; Keayla, 1999). Djibouti was chosen as a counter-example. It is a very small 'least developed state' that has never had any intellectual property system, no pre-existing expertise or institutions, and pharmaceutical patents have never had any existence in laws or debates up to now. Nor does it have any local pharmaceutical industry, which makes it slightly removed from international debates on pharmaceutical patents.

Several assumptions have persisted in the dominant literature in relation to the meaning of TRIPS and patents in 'developing states'. The causality of links between 'written creation' and 'legal action' has often been assumed (e.g. Fors, 2003; Matthews, 2004b). The fact that TRIPS would bring changes to the countries where it required pharmaceutical patents to be created was rarely questioned. Similarly, the fact that the role of pharmaceutical patents could be deduced from their legal status was rarely challenged.

This article aims to tackle some of these limits, by questioning the specific modes of actions of TRIPS and patents in Djibouti, by describing the range of network connections that create these specific modes of actions and by considering how these might impact on health and on disease in Djibouti.

EMPIRICAL FINDINGS: COMPLEXITY, CONTINUITY AND THE ROLE OF TRIPS AND PHARMACEUTICAL PATENTS IN DJIBOUTI

This study started with some assumptions made from existing literature, of TRIPS as a stable and rather simple legal object and of the action of pharmaceutical patents as depending on their official existence (e.g. Matthews, 2004b). In the case of Djibouti, a law on pharmaceutical patents would only exist because of TRIPS, and the two objects – TRIPS and pharmaceutical patents – were therefore expected to be overlapping. However, the empirical research

challenged these assumptions. Two aspects will be discussed here: the implementation of TRIPS on paper and through social networks and the action of pharmaceutical patents in spite of their lack of official existence.

*TRIPS AND 'LEGAL' ASSOCIATIONS: UNDERSTANDING THE ROLE
OF TRIPS AS PRESCRIPTION*

The first stage of the research analysed the action of TRIPS in social networks and questioned to what extent it mobilized actors. It also raised elements of network patterns that highlighted the differences between what could have been expected of TRIPS by its creators or by commentators and what it appeared to do on the ground in Djibouti. ANT was used as a method to question the modes of action of TRIPS, the various network ramifications and disruptions that it created and the relevant policy-network patterns in Djibouti that could explain the shaping of TRIPS.

ANT encourages researchers to question key moments and connections that make a particular tool act in a specific way. Before entering into the empirical findings, it is therefore useful to briefly recall the mechanisms of TRIPS, in order to understand better the contrasts between its expected and actual action in Djibouti. In relation to pharmaceutical patents, and as far as a country like Djibouti with no pre-existing patent law is concerned, TRIPS, once transferred from the WTO, should travel to the Ministry of Trade, which should create the appropriate set of legislation introducing pharmaceutical patents and help develop appropriate structures and expertise. Following this, and in order for TRIPS to carry the set of expected effects making patents active, it needs to modify the behaviour and action of various actors. More specifically, those importing and marketing drugs in the country need to respect the patent rights newly created. For them to do so, they should be made aware of the new limits set in their daily activities of what they can and cannot import or market. In order to be fully effective, a new system of pharmaceutical patents therefore necessitates the mobilization of actors not only in the ministry officially in charge of its implementation but also in the various systems that determine the local pharmaceutical market.

This section contrasts this expected story with the action of TRIPS in Djibouti. The story of TRIPS in Djibouti starts with its legal implementation in the Ministry of Trade, although this term is partly misleading and the networks followed only travelled through one particular office, in which one person officially worked and others occasionally visited. In this office, the traces left by TRIPS needed to be examined within the network where it officially entered Djibouti, following Latour's (2005) emphasis on the need to understand the social through the traces its circulation leaves – its (limited) material presence as a set of rules laid on paper, the traces of the WTO and others left in these materials, the words, thoughts and new concepts it left behind (predominantly expressed by the member of staff officially in charge of TRIPS).

When discussing the legal implementation of TRIPS and pharmaceutical patents, a first interview with a representative of the Ministry of Trade highlighted the various steps taken: 'We are now working on implementing patent law – because we have signed the TRIPS agreement. WIPO is offering us some advice when creating the law'.⁴ Also, 'We will also deal with questions of licences for making drugs and the questions raised by generic medicines will also be in there. But for the moment, we are only at a discussion stage – nothing is on paper yet' (Interview with a representative of the Ministry of Trade and Industry).

영국처럼 사법권
독히 (제정)을
도입하는 단계

The fact that TRIPS should trigger a set of reactions was not challenged at this stage: TRIPS had been signed; someone knew about it and was 'working on it'. There was no material outcome of this 'work' yet, no documents produced through it. Nevertheless, TRIPS had become part of discourses, generated some form of mobilization, and its inherent presence through what is the process of 'legal implementation' could be found through these. As such, and because Djibouti still had time to implement TRIPS,⁵ there was nothing unduly surprising about the fact that the material presence of TRIPS remained evasive.

The process of legal implementation of TRIPS as international law should not, however, be confused with the process of social implementation of TRIPS as a new policy, expected to redirect social actions and reshape actor-networks in a certain way, prescribed by its creators. This had to be questioned by following the networks where TRIPS was expected to produce changes of behaviour.

In the Ministry of Trade, discussions of the impact of TRIPS on actors' behaviour translated the projections that can be made of TRIPS on the basis of the expectations built around it. They were not, therefore, very different from what is often officially stated in relation to TRIPS and balanced the potential benefits of creating an intellectual property system with the relative weaknesses of TRIPS. The issue is more interesting when following the networks that TRIPS would need to mobilize to work as expected – including actors in other trade-related offices and in the public health field.

When questions about the role of the official implementation of TRIPS were extended within the trade field but outside of the office in charge, the potential limits to the enrolment of TRIPS as an active entity appeared: 'The way it works is that it doesn't change anything unless someone complains about what you are doing. We are so small that hopefully they won't complain!' (Interview with an actor working in the trade field). Similar limits appeared when following TRIPS in the public health field. For example,

This will not affect us in any way, for at least one good reason: international law doesn't mean anything here and nobody will change the way they work because of it. It is utopic to think it will have any effect, no one will be interested here. (Interview with a public health actor)

As far as my job here is concerned, TRIPS is not gonna change anything. I can't see anyone being interested here and I really can't imagine it changing anything. (Interview with a public health actor)

Overall, the agency of TRIPS as a set of prescriptive rules, outside the particular office where it is being legally implemented, appeared largely doubtful. It did not seem to mobilize any of the actor-networks in which it needs to be embedded to become active. This is particularly interesting given that TRIPS is often expected by commentators to be a key concern for developing countries and given the fact that, in the case of Djibouti, the amount of changes it requires is particularly important – structure, expertise, legislation and so on. The difficulty for creators of technologies of foreseeing the various networks in which their object will be enrolled and the potential gaps between expectation and actions are commonly studied in ANT (de Laet and Mol, 2000). In this particular case study, it translated into the need to investigate which aspects of the Djiboutian networks might have generated a gap between expectations built by the creators of TRIPS and the action of TRIPS in Djibouti.

EXPLAINING THE LACK OF SOCIAL ACTION OF TRIPS ANT emphasizes that the complexity of any network means that some dimensions and connections will remain hidden, even in the most thorough research process. To that extent, it is worth acknowledging that the empirical investigation of what made TRIPS partly dormant in Djibouti can by no means claim to be fully and definitely exhaustive (Law, 1994). However, some elements can be put forward to explain in part the specificities of Djiboutian networks as opposed to what might have been expected by creators or commentators of TRIPS.

The first specificity of the networks in Djibouti relates to the lack of expertise of most local actors:

I don't know much about this whole issue – only what I have read in the press.
(Interview with a public health actor)

The problem here is that no one has any kind of expertise here. There are a few pharmacists, but they won't understand what it is about. TRIPS, everyone talks about it but no one knows what it is about, so it just won't change anything!
(Interview with a public health actor)

This contrasts with writings and statements expecting a form of mobilization following TRIPS (Ford, 2003), whether in order to follow the prescriptions carried by TRIPS, or to oppose the resistance to TRIPS reported from a number of developing states.⁶ It can be explained partly by the fact that intellectual property is a completely new area in Djibouti. A minimal expertise is necessary, however, to ensure that some form of mobilization will be triggered by TRIPS; expertise that should in fact partly be generated by TRIPS and the system built around it. For some basic knowledge about TRIPS to develop in relevant fields, however, TRIPS, information about TRIPS, and intellectual property as an issue need to circulate across actor-networks in Djibouti, in trade and public health.

In Djibouti, a number of obstacles to this circulation appeared. The most central was the fact that current connections between the trade field, where

TRIPS was officially entering Djibouti and the public health field, where it needed to circulate for patents to become effective as official tools, were almost non-existent.⁷

Chronologically, this empirical research started with interviewing actors in the public health field. Connections were made quickly and informants knew each other enough to introduce me to whoever they thought might be a relevant contact. After having met most of the actors I wanted to talk to in the public health sector, I asked to meet 'their contacts' in the trade and industry field. The ease with which each informant had been able to introduce me to relevant colleagues made me – wrongly – assume that meeting someone in the trade-related field would be as simple. I truly expected that people creating rules on pharmaceutical imports for the Ministry of Health could introduce me to someone they knew working on imports more generally. Considering the current president of the *Chambre du Commerce* is also a pharmacist, I assumed the relations between the two sectors would be somehow facilitated. At least he would certainly know who works on trade and industry issues?

But, overall, it was impossible to find anyone in the Ministry of Health who could name anyone in the Ministry of Trade or the *Chambre du Commerce*. Relations between sectors were presented as rare:

I think the Minister here sometimes works with the Ministry of Finance to try to get some tax limitations for pharmaceuticals . . . but it is really sporadic. (Interview with a public health actor).

The difficulty of circulating between sectors is best illustrated by this example because of its centrality to this article, but appeared to be a characteristic of several other areas in Djibouti. It can seem surprising given the size of the country and the very small number of actors involved in any institution. It is also an illustration of the type of network dynamics that, if forgotten by creators of technologies or those commenting on the action of technologies, can result in their role being fundamentally different from what is expected from them.

Although an exhaustive explanation of this lack of circulation across policy fields in Djibouti is beyond the scope of this article, some elements can be raised that partly explain where channels/connections were missing. As a whole, the lack of circulation of TRIPS needed to be understood within the more general fragmentation of networks and institutions in Djibouti. This was due mainly to two factors: ethnic divisions that maintained some tensions and lack of communication in some instances on the one hand, and a persistent gap between nationals in some institutions and French cooperants on the other hand. The distribution of expertise, hierarchical division and underlying wariness were all part of what prevented some actors from efficiently liaising on new issues such as TRIPS.

However, more specific factors limiting the circulation of TRIPS appear when looking at the action of international organizations in this field in Djibouti. In existing literature, the action of international organizations, in particular the WHO, WTO and WIPO, are often portrayed as potentially

bridging the gaps between different sectors and areas relevant to pharmaceutical patents and participating in generating the relevant expertise. In the case of Djibouti, the impact of this action was questioned many times by informants:

The WHO has never talked to me about anything like that. We sometimes work with them on specific projects, but we don't see them very much. (Interview with a public health actor)

International advisors are really funny . . . they come here, stay here for a while, come to lots of cocktails, and meetings . . . and then after a few days or a few weeks, they go back home, and they send a report which is meant to be adapted to the specificities of the country . . . well, you just can't imagine how many times I received a report with in its introduction: 'Djibouti – Capital: Dakar or Nouakchott or something else' . . . They just take a report they had prepared for another country, and put 'Djibouti' in there. But what worked in Cambodia might not work here. It's a bit discouraging! (Interview with an actor working on trade)

When discussing whether information was received 'regularly' from the WTO:

Regularly . . . no! but sometimes! Once in a while! But things change so quickly that by the time we have been informed and decided to react, others have already moved on!' (Interview with a representative of the Ministry of Trade).

The system built internationally to increase expertise on TRIPS and allow TRIPS to develop its expected effects translated in Djibouti into a series of weak connections of little social impact, as comes across explicitly through these quotations. Once again, however, this is only one example of the type of obstacles that limited the circulation of TRIPS and it needs to be considered within broader ruptures across networks in Djibouti.

As a whole, TRIPS in Djibouti as a set of prescriptive rules appeared to be a rather weak social actor. Its story seems to indicate that pharmaceutical patents will not 'do anything' in Djibouti. The fact that they were officially non-existent prior to TRIPS could also suggest that they have never had any type of role in Djibouti. Nevertheless, within this context of apparent lack of mobilization and official 'legal emptiness', a closer look at the pharmaceutical market of Djibouti seemed to indicate that pharmaceutical patents were more strongly empowered than their official story might indicate.

EMBEDDING OF PHARMACEUTICAL PATENTS IN THE DRUGS USED IN DJIBOUTI

Even if pharmaceutical patents were only created because of TRIPS in Djibouti, investigating the impact of TRIPS in its pharmaceutical patents dimension turned out to be only one part of understanding the role of pharmaceutical patents in the country. This sub-section will explain how, in spite of their lack of official existence, pharmaceutical patents appeared as important actors in the pharmaceutical market of Djibouti. This will illustrate the

potential relevance of ANT as a method used to follow and highlight objects through their embedding in other connections (Latour, 2005) and will demonstrate how it can be used to comment on the complexity of relations between written legal rules and the social action of socio-legal objects, at unexpected levels.

PHARMACEUTICAL PATENTS AND LOCAL MARKET The concept of ‘hybrids’, defined in ANT as objects made of multiple human and non-human connections, can be useful to understand the nature of other multi-dimensional objects – including legal/technological hybrids such as ‘patents’ or ‘patented drugs’ (Latour, 1993). Patented drugs can therefore be understood as a particular type of hybrid, made up both from the complexity of pharmaceutical patents and of drugs. The social presence of pharmaceutical patents can be read within these hybrids and, when looking for traces left by pharmaceutical patents, it is essential to look at the drugs found in the system studied (Latour, 2005). Patents can then be understood as the gathering of heterogeneous connections that regulate and determine which drug will enter the local market. The necessity to understand interrelations, heterogeneity and complementarity as part of realities is evident from most ANT writings (Law and Singleton, 2000; Law, 2003; Latour, 2005).

In Djibouti, all public health actors interviewed originally dismissed pharmaceutical patents as something that was of no relevance to them. There is currently no law on patents and, as explained earlier, their introduction through TRIPS does not appear to mobilize political actors in any significant way. However, this official absence of patents from written law in Djibouti did not make patents/drugs any less present or any less active in the pharmaceutical market of the country.

Although patents were not present in words, texts or opinions, they were embedded in the range of medicines selected on medical prescriptions, within doctors’ and pharmacists’ strategies and patients’ requirements. If pharmaceutical patents are understood as more than their written/official dimension, if the hybrids that patented medicines represent are understood as keeping patents alive, patents were certainly a crucial part of the pharmaceutical network of Djibouti.

As a whole, generic medicines are not found in Djibouti, or only in very limited amounts:

In Djibouti, we don’t like or the doctors don’t prescribe, generic products yet so in the Republic of Djibouti we essentially don’t use generics. (Interview with a public health actor)

I own two pharmacies. One has about 90 per cent of foreign customers, the other 100 per cent locals. But the drugs are the same in both – no generics. (Interview with a pharmacist)

In the largest of the pharmacies, a drawer marked ‘generics’ contained a few boxes. Outside of this particular drawer, and with the very few exceptions of some ‘branded generics’, most drugs sold were the originally patented version

of a given product. The same was true in the different hospitals, with the limited exception of one hospital, where some generics could be found (only those that were not protected by patents anywhere). This exception was due to the semi-public structure of this hospital, which offered it both financial and practical independence and allowed the pharmacist to choose directly its sources of import. Other pharmacists and doctors confirmed that they did not deal with generics. Finally, doctors' prescriptions processed at private pharmacies equally contained lists of branded medications only. At the time of fieldwork, there was no drug – or at least no drug which I came across during my research – that would infringe patents if they were sold, for example, in Europe. Even drugs for which generic versions were legally sold in Europe only appeared as 'generics' in the one hospital mentioned earlier – and in one drawer. In fact, the system of Djibouti enrolled patents in a much more stable way than many European systems. In Europe, many drugs, once their patent has 'officially' run out, start appearing in new generic versions in pharmacies – new drugs *without* patents. But in Djibouti, except in one small hospital, patents socially exist independently from any written rule and any official, written temporal limit. Patents are not only what gives drugs market exclusivity for several years but they also determine which drug will or will not enter the territory with no pre-established time limit.

This is interesting given that written laws in Djibouti have never officially created patents, and even when they do, policy-makers explain that this would not mean that they will consider them prescriptive. The discrepancy between the written existence of law and its action in social networks adopted, in this case, an unexpected shape – a particular set of socio-legal objects appeared to have agency *without* being rooted in written rules. From a policy perspective, this is highly interesting, as it means that the Djiboutian pharmaceutical market, at the time of writing, was largely based on expensive drugs, out-of-reach for a substantial part of the population. It is problematic that a policy answer to the situation could not be helped in any way by the various types of recommendations made to developing states on how to limit the negative impacts of pharmaceutical patents on access to medications, since these are often drafted with the situation of states relying on existing written patent systems in mind. I have explored elsewhere the (limited) changes that are currently made to the Djiboutian pharmaceutical market, the extent to which there might be new prospects for generic medicines to enter the country and how these changes can be understood in the light of TRIPS (Cloatre, 2005). For the purpose of this article, however, it is useful to concentrate in more depth on analysing the set of social circumstances that explain why the pharmaceutical market of Djibouti is made of patented rather than generic medicines in the absence of any legal requirement that this be so.

The import system of Djibouti is predominantly private. Private pharmacists (two of them were established in Djibouti at the time of fieldwork) import almost every drug used in the country, including drugs used in most hospitals. When talking about private pharmacists with any governmental actor interviewed, a key element they emphasized was the complete lack of

power the government had over what pharmacists chose to import. Pharmacists essentially act as commercial actors, and remain motivated by what will be most convenient, practical, or financially rewarding, rather than by governmental policies. Because of the predominant role played by private pharmacists in Djibouti, as compared to complementary sources, this section will be used to briefly explain how patents become embedded in the imports made by private pharmacists.

EMBEDDING OF PATENTS/DRUGS IN THE IMPORT SYSTEM OF DJIBOUTI This section questions the range of connections that make patents active and empowered. It discusses how these connections have become stable enough – ‘translated into a stable network’ (Law, 1992) – for the import network of Djibouti to exclude any generic version of patented drugs and exclude *any* generic in a predominant part of the market. Methodologically, these connections were studied by following the various threads provided by informants, in order to draw the more general picture of a network where patents have become enabled and active (Latour, 2005).

The first direction taken by informants, and the first set of connections relevant to the network under scrutiny, associated tightly private pharmacists to multinational companies, mainly based in France. This was opposed to an apparent absence of links between pharmacists and generic companies. Several elements were presented as relevant to understanding this pattern. The first one related to channels of information. Pharmacists explained that information from generic producers did not reach them, while multinational companies were proactive in making their products known in Djibouti. Materials of various kinds played a part in stabilizing these networks of information. Leaflets and visits from representatives were mentioned as key strategies. The fact that Sanofi even had a permanent local representative in charge of organizing monthly meetings with health professionals where specific products would be displayed was also given as an example of information strategy. In addition to this, one pharmacist explained that they attend the Pharmagora meetings held in Paris every year as a way of establishing direct contact with companies that had been in touch through the profusion of leaflets, posters, advertising pens and writing pads that were sent over by pharmaceutical companies mostly from France.

However, in order to illustrate in more detail the network through which patents have become embedded, other connections need to be drawn. These have to be understood within other overlapping networks and, in particular, the general connections between Djibouti and other countries (Law, 1999c). The practicalities involved in materials travelling to Djibouti were highlighted as partly explaining the strengths of links between pharmacists in Djibouti and some multinationals. Shipping drugs to Djibouti was said to be particularly lengthy and, once importers were convinced that the medication they ordered would reach them as planned, they tended to remain with the same providers. Informants thus emphasized several times that they tend to keep the same providers because ‘at least we know it works!’.

Networks of information and travel explain why large companies are privileged over smaller generic producers by pharmacists in Djibouti. However, the fact that even 'branded generics' are very rarely found in Djibouti is not explained by this link between private importers and a few large companies – as companies such as Merck, for example, produce their own branded generics. The role of doctors in stabilizing key connections between patients, diseases and patented drugs explains this missing link.

This role itself is embedded within wider persistent patterns inherited from the colonial relations established between France and Djibouti. French doctors represent an important part of the public health network of Djibouti and their share in the prescribing system is significant enough for their behaviour to influence the actions of pharmacists. Research in France has demonstrated that French doctors tend to prescribe branded medication rather than refer to the generic name of medicines (Kapfere, 1997; Larrieu and Houin, 2001). This is linked in France both to medical training and to the shared responsibilities between doctors and pharmacists – pharmacists being allowed to replace any branded medication prescribed by its generic version. Similar patterns of doctors' behaviour could be identified in Djibouti: 'That's what we've been taught at uni and maybe for younger doctors it is different, but if I need to prescribe Prozac or Primperan I can't tell you the generic name and I don't have time to go and look it up' (Interview with a French doctor).

Patents are therefore embedded in most medical prescriptions in Djibouti. Only positive decisions by pharmacists to replace the branded drugs prescribed by their generic version could challenge this. However, pharmacists suggested that they would not consider doing so, essentially because patients would refuse anything else than what was written on the prescription: 'Patients don't want generics. They want what's written on the paper' (Interview with a pharmacist).

The action of pharmaceutical patents in the market of Djibouti is co-dependent on a variety of relations, overlapping with networks such as public health system, transports or neo-colonialism, among others. The weakness of some of these connections are relevant to the patterns of empowerment of patents, such as the weakness of shipping networks and of material connections between Djibouti and other places where drugs are being produced.⁸ Next to these restrained networks, several stable systems have developed, built around two private pharmacists and their shops, obligatory points of passage (Callon, 1986) for most drugs entering Djibouti, which are entangled in sets of strong and well-run connections that they expect to remain stable and foresee as unlikely to crumble. The connections between medicine in Djibouti and training in France, drugs in Djibouti and multinational companies based in France – themselves resulting from multiple involvements and exclusions – all resulted in sets of associations through which routines that can be understood as emerging from pharmaceutical patents, as official tools in France were repeated and embedded in habits of actors in Djibouti. In addition to this, all these patterns only took this shape because, at the time of fieldwork, competing networks directed by defined policies

towards poorer people who would have needed access to cheaper medication were only at an initial stage.

Overall, although officially pharmaceutical patents are non-existent in Djibouti, these connections and associations resulted in them being fully embedded in drugs, prescriptions, routines and actions relevant to the pharmaceutical network of Djibouti. Bypassing their non-existence in written national law in Djibouti, this tool that officially emerged on paper in other countries, supposedly for other countries, has become centrally realized through the connections lived and relived by private importers in Djibouti.

CONCLUSION

As a whole, the story of TRIPS and pharmaceutical patents in Djibouti reflected many discrepancies between expectations and findings, official stories and social action and between the letter of socio-legal objects and their action in social networks. Although a broad evaluation of the merits and limits of ANT as a method is partly beyond the scope of this article, a few words should be said on what the method brought to the aspects of the research presented in this paper. It should be noted, however, that this article only explores some aspects of the original research, and the merits of ANT for this project can therefore only be evaluated in part. This research has also demonstrated some of the limits of ANT when relied upon in socio-legal research, and this conclusion aims to discuss these.

The relevance of ANT needs to be evaluated in relation to the two main findings presented here. In terms of analysing the impact of TRIPS in mobilizing actors within policy-networks in Djibouti, it first has to be underlined that the type of analysis provided remains closely related to existing approaches more traditionally associated with the study of policies and regulations – such as implementation studies. In particular, the empirical bottom-up approach used in this case study, starting from the point of view of recipients before moving back to compare the potential discrepancies with the aim of creators, relates to similar trends in implementation studies emphasizing the complexity of implementation processes and the set of transformation that any policy will potentially go through (Berman, 1978; Goggin et al., 1990).

Nevertheless, ANT's emphasis on the need to follow connections, actors, and to always look into the various constituents of each network remained, in my view, essential in helping draft the relevant connections built (or missing) around TRIPS and relevant policy networks. The notions of heterogeneity, symmetry and materiality are also certainly useful concepts in helping move what can be considered the centre of a particular network, or origin of a particular action, from human to non-human – with the limit, discussed later, of the role of immaterials within these networks.

The second part of this research questioned to what extent ANT can be used as an approach to study absent socio-legal objects – in this case the role of patents in spite of their lack of official existence. Once again, ANT's

emphasis on continuity and on the need to ground findings in network effects rather than the nature of particular objects allowed non-official objects to be observed and seen as powerful. In this example, the action of pharmaceutical patents was observed aside of their official role and through their base in drugs. Similarly, it became an essential aim of the research to empirically follow the networks that embedded them. These networks could include fully active materials (patents, but also ships, leaflets, prescriptions) whose action was related to, but interdependent rather than passively dependent on, human action. The networks to be traced were drawn from patents and around patents, in order to understand how they managed to act.

One of the questions that could be asked, however, is whether this approach might in fact have resulted in too much emphasis being put on patents. One of the difficulties is to know how to draw conclusions from the *presence* of patents within drugs towards the *action* of pharmaceutical patents as socio-legal objects/regulatory tools. The difficulty comes in part from the fact that the detail of their interaction with various actors and materials is potentially more difficult to observe than that of material objects which ANT has been developed to study. To some extent, once 'melted into' drugs, the specific role of patent becomes difficult to identify and follow empirically. The answer would be that this type of analysis is more about presence than about causality; that it is not about individual action but about network/hybrid action. In this example, once patents are understood as being partly found in patented drugs, their role as expressed through the role of these hybrids is what the research should observe.

Nevertheless, this highlights what might be the most central criticism that can be made of ANT on the basis of this research: the extent to which ANT offers sufficient scope to theorize the *immaterial*. The likelihood of legal objects to be reshaped through words and intellectual concepts in legal practice might differ from the changes technical or scientific objects go through in their own practice, in laboratories or in society. The example of the embedding of patents in patented drugs also shows that socio-legal objects can become elusive and while their presence remains identifiable – and while ANT is actually a useful approach to conceptualize their presence – their modes of actions remain difficult to seize. On the one hand, the emphasis of ANT on non-human materials is useful because looking for socio-legal objects can involve, as in this case study, looking within materials and within heterogeneous networks; on the other hand, there remains some difficulty in observing the action of immaterial objects embodied within material networks to which ANT might not yet offer a full solution. While it does offer a way to draw networks and patterns in which these objects can found, the strengths it has developed to analyse modes of actions of material objects might still need further elaboration to be transferred to more elusive objects.

Overall, and in spite of the benefits that ANT brought to this research in terms of offering an encompassing and flexible method to reflect on the subject matter, it also appears from this analysis that its transfer to socio-legal research might require further theorizing, best achieved through *practice*.

While Latour (2005) stated that legal objects act in a 'legal way', this might need further elaboration. It appeared here that the definition of the modes of action of so-called 'legal' objects could not be pre-defined and certainly not perceived as acting in an inherently different, independent or unique way that could be isolated from other social connections. The 'legal' way needs to be understood here as a socio-legal range of connections fully embedded in various types of practices and networks. Nevertheless, the elusive nature of socio-legal objects might require that additional thought be given to how the modes of action of these immaterial tools can be observed and analysed in *practice*.

NOTES

I would like to thank Tamara Hervey, David Fraser, Dave Cowan as well as the anonymous reviewers for *Social & Legal Studies* for their constructive comments on earlier versions of this article. The support of the ESRC for this research is gratefully acknowledged.

1. The term 'socio-legal' is also used as a way to emphasize the continuity between the two fields. The term 'object' should not be understood as pre-determining any form of 'passivity'.
2. For a full text of TRIPS see http://www.wto.int/english/docs_e/legal_e/27-trips.doc
3. Versions of drugs others than the patented version – usually significantly cheaper (Dumoulin, 2000).
4. Anonymity is a particularly sensitive issue when working in a country like Djibouti, given that the small number of actors involved in each institution makes it very easy to retrace their identity, even with limited information. For this reason, only basic information on interviewees will be provided throughout this article.
5. Until 2016 for pharmaceutical patents to be fully in place, although other measures had to be implemented since 2000.
6. The term 'prescription' reflects the idea of anticipated mobilization (Akrich and Latour, 1992). It is used here to suggest that TRIPS is expected to carry effects anticipated to mobilize actors in a certain pre-defined way by its creators.
7. This is not fundamentally different from findings made in other contexts (Barry, 2002).
8. Analyses of 'missing links' are not the most central emphasis of ANT, but can be found, for example, in (Star, 1991).

REFERENCES

- Abbott, Frederick M. (2002) 'The Doha Declaration on the TRIPS Agreement and Public Health: Lighting a Dark Corner at the WTO', *Journal of International Economic Law* 5(2): 469–505.
- Abbott, Frederick M. (2005) 'Managing the Hydra: The Herculean Task of Ensuring Access to Essential Medicines', pp. 393–425 in Keith E. Maskus and Jerome H. Reichman (eds) *International Public Goods and Transfer of Technology under a Globalized Intellectual Property System*. Cambridge: Cambridge University Press.

- Akrich, Madeleine and Bruno Latour (1992) 'A Convenient Vocabulary for the Semiotics of Human and Non-human Actors', pp. 205–24 in Bijker Wiebe and John Law (eds) *Shaping Technology, Building Society: Studies in Sociotechnological Change*. Cambridge, MA: MIT Press.
- Barry, Andrew (2002) 'In the Middle of the Network', pp. 142–65 in John Law and Anne-Marie Mol (eds) *Complexities: Social Studies of Knowledge and Practice*. Durham, NC: Duke University Press.
- Berman, Paul (1978) 'The Study of Macro and Micro Implementation', *Public Policy* 27: 157–84.
- Braune, Fernando and Naira J. Menezes (1998) 'The Patentability of Chemical, Biochemical, Pharmaceutical and Biotechnological Inventions in Brazil', *Patent World* 108: 46–8.
- Callon, Michel (1986) 'Some Elements of a Sociology of Translation: Domestication of the Scallops and the Fishermen of Saint-Brieuc Bay', pp. 196–233 in John Law (ed.) *Power, Action and Belief: A New Sociology of Knowledge?* London: Routledge and Kegan Paul.
- Cloatre, Emilie (2005) *Actor-Network Theory and Socio-legal Objects: Analysing TRIPS and Pharmaceutical Patents in the Republic of Djibouti*. PhD Thesis, University of Nottingham.
- Correa, Carlos M. (2000) *Intellectual Property Rights, the WTO and Developing Countries: The TRIPS Agreement and Policy Options*. London: Zed Books.
- de Laet, Marianne and Anne-Marie Mol (2000) 'The Zimbabwe Bush Pump: Mechanics of a Fluid Technology', *Social Studies of Science* 30: 225–63.
- Draho, Peter (2002) 'Negotiating Intellectual Property Rights, Between Coercion and Dialogue', pp. 161–82 in Peter Draho and Ruth Mayne (eds) *Global Intellectual Property Rights: Knowledge, Access and Development*. London: Palgrave.
- Dumoulin, Jérôme (2000) 'Les Brevets et le Prix des Médicaments', *Revue Internationale de Droit Economique*: 45–69.
- Ford, Nathan (2003) 'Patents, Access to Medicines and the Role of Non-governmental Organisations', *Journal of Generic Medicines* 1(2): 137–45.
- Goggin, Malcolm L., Ann Bowman, James Lester and Laurence O'Toole, (1990) *Implementation Theory and Practice: Towards a Third Generation*. Upper Saddle River, NJ: Scott Foresman.
- Halbert, Deborah (2002) 'Moralized Discourses: South Africa's Intellectual Property Fight for Access to AIDS Drugs', *Seattle Journal for Social Justice* 1: 257–88.
- Henderson, Elizabeth (1997) 'TRIPS and the Third World: The Example of Pharmaceutical Patents in India', *European Intellectual Property Review* 19(11): 651–63.
- Lanjouw, Jean O. (1998) *The Introduction of Pharmaceutical Product Patent in India: Heartless Exploitation of the Poor and Suffering?* Yale University, Economic Growth Centre.
- Latour, Bruno (1979) *Laboratory Life: The Social Construction of Scientific Facts*. Los Angeles: SAGE.
- Latour, Bruno (1987) *Science in Action: How to Follow Scientists and Engineers Through Society*. Milton Keynes: Open University Press.
- Latour, Bruno (1988) *The Pasteurization of France*. Cambridge, MA: Harvard University Press.
- Latour, Bruno (1993) *We Have Never Been Modern*. Brighton: Harvester Wheatsheaf.
- Latour, Bruno (1999) 'On Recalling ANT', pp. 15–25 in John Law and John Hassard (eds) *Actor-network Theory and After*. Oxford: Blackwell.
- Latour, Bruno (2005) *Reassembling the Social: An Introduction to Actor-network Theory*. Oxford: Oxford University Press.
- Law, John (1992) 'Notes on the Theory of Actor-network: Ordering, Strategy and Heterogeneity', *Systems Practice* 5: 379–93.

- Law, John (1994) *Organizing Modernity*. Oxford: Blackwell.
- Law, John (1999a) *Objects, Spaces and Others*. Lancaster: Lancaster University.
- Law, John (1999b) 'After ANT: Complexity, Naming and Topology', pp. 1–14 in John Law and John Hassard (eds) *Actor-network Theory and After*. Oxford: Blackwell.
- Law, John (1999c) *Materialities, Spatialities, Globalities*. Lancaster: Lancaster University.
- Law, John (2000) *Ladbroke Grove, or How to Think about Failing Systems*. Lancaster: Lancaster University.
- Law, John (2002) *Method, Complexity and the Baroque*. Lancaster: Lancaster University.
- Law, John (2003) *And If the Global Were Small and Non-coherent? Method, Complexity and the Baroque*. Lancaster: Lancaster University.
- Law, John and Vicky Singleton (2000) *This Is Not an Object*. Lancaster: Lancaster University.
- Mansfield, Edwin (1994) 'Intellectual Protection, Foreign Direct Investment and Technology Transfer', *Journal of International Economics* 16: 205–26.
- Maskus, Keith E. (1998) 'The Role of Intellectual Property in Encouraging Foreign Direct Investment and Technology Transfer', *Duke Journal of Comparative and International Law* 9: 109–61.
- Matthews, Duncan (2004a) 'Is History Repeating Itself? The Outcome of Negotiations on Access to Medicines, the HIV/AIDS Pandemic and Intellectual Property Rights in the World Trade Organisation', *Law, Social Justice and Global Development Journal* 1. Available at: http://www2.warwick.ac.uk/fac/soc/law/elj/lgd/2004_1/matthews/
- Matthews, Duncan (2004b) 'WTO Decision on Implementation of Paragraph 6 of the Doha Declaration on the TRIPS Agreement and Public Health: A Solution to the Access to Essential Medicines Problem', *Journal of International Economic Law* 7(1): 73–107.
- Mayer, Christopher S. (1998) 'The Brazilian Pharmaceutical Industry Goes Walking from Ipanema to Prosperity: Will the New Intellectual Property Law Spur Domestic Economy?', *Temple International and Comparative Law Journal* 12: 377–89.
- Muennich, Franz E. (2000) 'Les Brevets Pharmaceutiques et l'Accès aux Médicaments', *Revue Internationale de Droit Economique*: 71–81.
- Nogues, Julio (1993) 'Social Costs and Benefits of Introducing Patent Protection for Pharmaceutical Drugs in Developing Countries', *The Developing Economies* 31(1): 24–53.
- Otten, Adrian (2000) 'Les Brevets Couvrant les Produits Pharmaceutiques et l'Accord sur les ADPIC', *Revue Internationale de Droit Economique*: 161–6.
- Primo, Braga, Carsten Fink and Claudia P. Sepulveda (2000) 'Intellectual Property Rights and Economic Development', World Bank Discussion Paper no. 412. Available at: <http://www.bvindecopi.gob.pe/colec/cbraga.pdf>
- Scherer, F. M. and Jayashree Watal (2002) 'Post-TRIPS Options for Access to Patented Medicines in Developing Nations', *Journal of International Economic Law* 5(4): 313–19.
- Star, Susan Leigh (1991) 'Power, Technologies and the Phenomenology of Conventions: On Being Allergic to Onions', pp. 26–56 in John Law (ed.) *A Sociology of Monsters*. London: Routledge.
- Tully James (2006) 'A New Kind of Europe? Democratic Integration in the European Union', Constitutionalism WebPapers, ConWEB 4/2006. Available at: <http://www.qub.ac.uk/schools/SchoolofPoliticsInternationalStudiesandPhilosophy/FileStore/ConWEBFiles/Filetoupload,38375,en.pdf>
- Velasquez, German (2000) 'Médicaments Essentiels et Mondialisation', *Revue Internationale de Droit Economique*: Vol. 1 38–44.